

1. Information about the indicator	
Goal	End hunger, achieve food security and improved nutrition and promote sustainable agriculture
Task	By 2020, maintain the genetic diversity of seeds, cultivated plants and farmed and domesticated animals and their related wild species, including through soundly managed and diversified seed and plant banks at the national, regional and international levels, and promote access to and fair and equitable sharing of benefits arising from the utilization of genetic resources and associated traditional knowledge, as internationally agreed
The indicator	Number of plant and animal genetic resources for food and agriculture secured in either medium- or long-term conservation facilities
2. Information about the organization	
Organization	Ministry of Agriculture of the Republic of Kazakhstan, NAO "National Agrarian Scientific and Educational Center"
3. Definitions and concepts	
Definition	The indicator is defined as the number (units, doses) of available genetic resources of agricultural plants and animal breeds in the Republic of Kazakhstan and the amount necessary for conservation, replenishment and documentation in order to study useful properties and their use in order to obtain scientific knowledge and develop commercial products.
Basic concepts:	Genetic resources of agricultural crops and animal breeds mean genetic material of actual or potential value. Biological diversity refers to the variability of living organisms from all sources, including, but not limited to, terrestrial, marine and other aquatic ecosystems, and the ecological complexes of which they are part; this concept includes diversity within species, between species, and ecosystem diversity. Biological resources include genetic resources, organisms or parts thereof, populations, or any other biotic components of ecosystems that have actual or potential utility or value to humanity. Ex-situ conservation means the conservation of components of biological diversity outside their natural habitats. In-situ conditions mean the conditions in which genetic resources exist within ecosystems and natural habitats, and, in the case of domesticated or cultivated species, in the environment in which they have acquired their distinctive characteristics. (Source:Convention on Biological Diversity, 1992).
Justification and interpretation	The obvious narrowness of the genetic base, the lack of genetic diversity with a higher threshold level of sustainability and stability requires research that will enhance and expand the adaptation of the agro-industrial complex of the republic to the new changing requirements of agronomy, nutrition and climate. The necessity of studying the genetic variability of agrobiodiversity according to the characteristics important for direct use as a starting material for breeding new varieties is traced. Monitoring of viability and regular regeneration of seed collections with a critical germination rate (< 50%) is the basis for safe storage of the collected gene pool. The main argument in favor of ex-situ conservation: it is suitable for a large number of species; it guarantees the control of the preservation of the sample and the possibility of constant accounting and control over the progress of the material. Kazakhstan has adopted and signed a number of international documents on the conservation and development of PGRFA, the most important of which are the Convention on Biological Diversity (1992) and the Nagoya Protocol (2015). Collection, systematization, accumulation, storage, updating, modification,

	use, dissemination, blocking and destruction of data on introduction, study and storage form the basis of information technology (IT) in the field of biodiversity protection, the importance of which is reflected in a number of international documents. Documentation is the basis for centralized management, data processing capabilities, and the creation of a single database for user operational availability. There are technical and software problems that complicate the effective management of the PGRFA of the Republic of Kazakhstan both at the local level and in the process of international exchange of plant samples. Gene pool activities require a long time period to carry out a large number of diverse activities related to replenishing collections with new samples, studying and identifying sources of traits valuable for breeding, developing an electronic collection documentation system, creating a system for reliable storage and maintaining high viability of seed samples. With the increasing intensification of animal husbandry, the use of artificial insemination and embryo transplantation, the issue of the irretrievable loss of many biologically valuable genotypes of farm animals and poultry is becoming more acute. The focus of breeding on achieving maximum productivity of animals, increasing the economic efficiency of livestock production, can lead to the complete extinction of many breeds and even some species of animals. In Kazakhstan, as a result of the long-term application of effective breeding methods, isolated populations and animal breeds have been developed regionally, as the Kalmyk beef breed of cattle, the Semirechenskaya breed and the Aksai black-and-white group of pigs, the mountain population of Galloway cattle, the Alatau and Aulietian dairy and meat breeds, etc. All of these genotypes are inferior to factory breeds in terms of productivity, but they have clear advantages in terms of endurance, adaptability to feeding and maintenance conditions, and resistance to diseases. The preservation of small and endangered local breeds and genotypes is primarily determined by breeding needs for them, as a source of genetic variability in the creation of new and improved existing breeds. Measures to preserve the gene pool will be implemented after carrying out work on the examination of genetic reserves and the development of scientifically sound recommendations for the conservation, development and management of genetic resources of animal husbandry in Kazakhstan.
4. Data sources and collection methods	
Data sources Data	If budgetary funding is allocated for scientific programs, the source will be the reporting data of the research institutes of the non-profit joint-stock company National Agrarian Scientific and Educational Center
collection methods	Field trips, electronic Databases, screening of agrobiodiversity, storage – analysis of viability and restoration of samples of seed genebanks with a critical level of viability, use of data from the Information and Analytical System in animal husbandry, identification of farm animals.
Unit of measurement	The amount of genetic resources of plant and animal origin in units, doses.
Periodicity	Annually
5. Calculation method and other methodological considerations	
Calculation method:	2021-2023 years Agricultural crops: Mobilization of national and global genetic resources of agricultural crops ≈

500 samples:

- 50 samples - expedition collection;
- 300 - international variety testing;
- 150- exchange between the National Research University of the Republic of Kazakhstan and abroad.

Electronic Databases of ≈ 2000 samples:

- mobilization – 500 units;
- monitoring – 500 units;
- conservation - 1000 units.

Screening of agrobiodiversity ≈ 2000 mod.:

- phenotyping - 2000 mod
- genotyping – 100 mod.

Storage of ≈ 2000 samples:

- determination of viability – 1000 units.

-restoration of samples with a critical level of viability ($\geq 50\%$) - 1000 samples.

Transfer of ≈ 300 samples (donors and sources of valuable traits, hybrid material):

- 2021 – 100 units;
- 2022 year – 100 rubles;
- 2023 – 100 units .

Farm animals:

The following standards are recommended for creating a spermobank::

- in cattle breeding, semen frozen in liquid nitrogen is stored for long-term storage from at least 10 breeding bulls representing five leading factory (genealogical) lines.

At least 400 doses per bull must be frozen.;

- in sheep breeding, the sperm of 10-12 sheep from 5-6 lines is frozen;
- in pig breeding – 10-12 boars, the main 5-6 lines;
- there are at least 50 roosters in poultry farming.

Recommended organizational forms of gene pool conservation:

Destination	Name	Location, breed, genotype	Conditions
Relic farms	Preservation of the genetic reserve	Endangered native breeds and populations	Purebred breeding with an outbred type of selection
Gene pool management	Preservation of the genetic reserve	Dwindling herds of domestic breeds.	Purebred breeding with an outbred type of selection.
Gene pool storage of sperm from producers-available breeds, types and lines	Conservation and application in breeding of the gene pool of valuable breeds	At leading research institutes and breeding centers	Modern facilities and cryogenic equipment, a reasonable structure of the sperm stock by breed and

	species. Calculations show that in general, in order to preserve genetic resources in Kazakhstan, based on bred and cultivated breeds, 80,000 doses of meat, dairy and combined breeds must be reserved for cattle breeding, sheep and goats – 90,000 doses from breeds, pigs – 14,400 doses from 4 breeds, bees – 3,000 doses from 3 breeds, in particular There are 45,000 doses of 15 breeds in horse breeding, as well as the creation of gene pools for poultry, 2 types of camels, as well as marals and spotted deer.
Comments and restrictions:	Risks may be associated with force majeure circumstances of a natural or man-made nature, including quarantine measures related to a pandemic of particularly dangerous infections.
6. Data availability and disaggregation	
Data availability and gaps:	Data is not published, upon request
The level of disaggregation:	The data are disaggregated by species and variety composition of agricultural crops, species and breeds of farm animals
7. Comparability with international data/standards	The amount of genetic resources of plant and zoological origin intended for food production and agriculture, which are stored at special facilities for either medium-term or long-term storage. According to the materials of the inventory conducted in 2025 (NAO NANOC Order No. 04-02/5 dated January 27, 2025 on inventory), 63376.0 samples of the most important agricultural crops, including 1704 samples of wild relatives of agricultural crops, are stored in 15 subsidiaries of the NAO "National Agrarian Scientific and Educational Center". Pursuant to the Decree of the President of the Republic of Kazakhstan No. 1025 dated March 17, 2015 on Kazakhstan's accession to the Nagoya Protocol on Regulation of Access to Genetic Resources and Fair and Equitable Benefit Sharing from their Application to the Convention on Biological Diversity (Decree of the President of the Republic of Kazakhstan No. 1025 dated March 17, 2015) and Resolution of the Government of the Republic of Kazakhstan No. 222 On March 16, 2023, KazNII ZiR LLP was designated as a control point for monitoring and increasing transparency in the use of genetic resources., as well as the preservation and development of the gene pool of highly valuable varieties of agricultural plants (Order of the Ministry of Agriculture of the Republic of Kazakhstan No. 186 dated May 17, 2023). The comparability of research data on the genetic resources of the Republic of Kazakhstan with international data / standards will be carried out using international methods and descriptors.: Mobilisation 1. Painting K. Introduction to Collecting, IPGRI, - Rome, 1996. – 39 pp. Monitoring 2. Scientific Management of germplasm: characterization, evaluation and enhancement, - IPGRI. – 1998. 3. Prescott J. M., Burkett P.A., Sari E.E. and others. A guide for the field definition of “Wheat diseases and pests” // GTC-CIMMYT. - 2002. - 135 p. Stubbs R.W., Prescott J.M., Saari E.E. and Dubin H.J. 1986. 'Cereal Disease Methodology Manual.' (CIMMYT: Mexico.) (Cobbscale) Documentation 4. “Instructions for the Management and Reporting the

	Results”//CIMMYT/ICARDA/OSU. – 2000. - 14 pp. 5. Painting K.A., Perry M.C., Denning R.A., Ayad W.G. // Guidebook for Genetic Resources Documentation. – IPGRI. -1995. Keeping 6. Engels J.M.M. Genebankstandarts. - FAO/IPGRI. – 1993. -13 pp. The gradual introduction of an International Classification can improve the quality and comparability of national and international data. 7. Altukhov Yu.P. Intraspecific genetic diversity: monitoring and principles of conservation / Yu.P.Altukhov // Genetics. 1995. - No. 10 - S. 13331357. 8. Frenkel, O.H. Group Genetic Resources for Today and Tomorrow / International Biological Program 2. / O.H. Frenkel, J.G. Hawkes Cambridge University Press - Cambridge, 1975. 162 p. 9. Management of biological diversity in agricultural ecosystems// Bioversityinternational publications. Rome 2012. C-476. 10. The state of the world's animal genetic resources in the field of food and agriculture// FAO publications. Rome 2010. C-512. Frankel, O.H. Conservation and Evolution / O.H. Frankel, M.E. Soul - Cambridge University Press. Cambridge, 1981. - 68 p. 12. Harris R.B., Genetically effective population size of Large mammals: an assessment of estimators / R.B. Harris, F.W. Allendorf // Coserv. Boil. 1989. - №3-P.181-191
8. References and documentation	https://online.zakon.kz/Document/?doc_id=30063141#pos=3;-260 http://www.fao.org/cgrfa/topics/plants/ru/www.wipo.int/edocs/mdocs/tk/ru/wipo_grtkf_ic.../wipo_grtkf_ic_38_inf_7.docx; https://www.cbd.int/doc/c/f495/49a7/.../sbstta-22-12-ru.docx https://www.cbd.int/doc/world/kz/kz-nr-05-ru.pdf http://kaf.kz/download/dev_apk_2020.pdf